Automated Blood Group Detection Using Computer Vision and Transfer Learning on Mobile-Optimized Deep Neural Networks

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***Abstract*—Accurate and rapid blood group identification is a critical step in transfusion medicine, emergency care, and surgical procedures. Conventional haemagglutination-based methods, while reliable, are time-consuming, operator-dependent, and susceptible to human error. This paper presents an automated, AI-driven blood group detection system that combines COCO-format bounding-box annotation, computer-vision region extraction, and MobileNetV2 transfer learning to identify ABO and Rh(D) blood groups directly from photographs of standard blood-typing cards. The proposed pipeline extracts three reagent zones (Anti-A, Anti-B, Anti-D) from a single card image, classifies the clumping intensity in each zone as *Strong Clumping*, *Medium Clumping*, or *No Clumping*, and applies deterministic ABO + Rh logic to output the final blood group. The system is trained on a comprehensive annotated dataset comprising 6,456 training images, 997 validation images, and 383 test images, achieving approximately 94–95 % zone-level classification accuracy with a macro-average F1-score of 0.947. The lightweight MobileNetV2 backbone (3.4 M parameters) enables deployment on resource-constrained devices such as smartphones, making the system viable for point-of-care use in low-resource healthcare settings. Limitations, including annotation dependency and sensitivity to lighting, are discussed alongside a roadmap for future real-time and annotation-free deployment.**

***Index Terms*—blood group detection, transfer learning, MobileNetV2, COCO annotation, computer vision, haemagglutination, point-of-care diagnostics**

I. Introduction

A. Problem Statement

Blood transfusion incompatibility is one of the leading causes of transfusion-related fatalities worldwide. The ABO and Rh(D) blood group systems are the two most clinically significant antigen systems; mismatching in either system can trigger life-threatening haemolytic reactions [1]. Manual typing by trained laboratory staff using glass-slide or tube haemagglutination assays remains the gold standard, but requires skilled personnel, controlled laboratory conditions, adequate time, and consistent lighting—resources that are not always available in emergency rooms, military field hospitals, or rural clinics.

B. Motivation

The proliferation of high-resolution smartphone cameras and the availability of lightweight deep learning frameworks (TensorFlow Lite, ONNX) create an opportunity to democratise blood group typing. A system that can classify a blood group from a photograph of a standard disposable typing card—taken under ordinary ambient light—would dramatically reduce the diagnostic bottleneck at the point of care.

C. Contributions

This work makes the following contributions:

1) A modular, end-to-end Python pipeline that extracts reagent regions from blood card images using COCO bounding-box annotations, classifies clumping intensity with MobileNetV2, and applies deterministic logic to determine ABO + Rh blood group.
2) A comprehensive annotated dataset comprising 6,456 training images, 997 validation images, and 383 test images (total 7,836 images) covering blood groups A, B, AB, and O, achieving 94–95 % zone-level accuracy and a macro-average F1-score of 0.947 via MobileNetV2 transfer learning from ImageNet.
3) A clean, production-ready codebase with error handling, modular functions, and CLI support for training and inference.
4) A quantitative evaluation framework (accuracy, precision, recall, F1, confusion matrix) and a detailed roadmap for deployment on Android devices via TensorFlow Lite.

II. Literature Review

A. Traditional Blood Group Detection Methods

Haemagglutination tests—where red blood cells mixed with specific antisera clump together (agglutinate) if the corresponding antigen is present—have been used since Landsteiner's discovery of the ABO system in 1901 [2]. Three standard formats exist: glass-slide, tube, and microplate (gel card) agglutination. Each requires trained technicians to visually interpret reaction strength, introducing inter-observer variability of 2–5 % [3].

B. Limitations of Manual Methods

Table I summarises the key limitations of conventional manual methods.

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TABLE I
LIMITATIONS OF MANUAL BLOOD TYPING METHODS

Limitation	Impact
Operator skill dependency	Variable accuracy between technicians
Reaction-strength ambiguity	Misclassification of weak agglutination
Throughput constraints	Unsuitable for mass-casualty events
No digital record	Manual transcription errors
Cost	Gel-card systems are expensive

C. Existing Computer-Vision Approaches

Early automated approaches relied on image-processing heuristics: Saif *et al.* [4] used HSV colour thresholding and blob detection to segment agglutinated regions on glass slides, achieving 88 % accuracy. Khan *et al.* [5] employed a support-vector machine (SVM) on hand-crafted texture features extracted from the Grey-Level Co-occurrence Matrix (GLCM), reporting 91.3 % accuracy on a controlled laboratory dataset.

The advent of convolutional neural networks (CNNs) substantially raised performance. Yildiz *et al.* [6] applied ResNet-50 directly to cropped reagent images, achieving 93.7 % accuracy. Ahmed *et al.* [7] used VGG-16 with data augmentation, reporting 94.2 % on a 1200-image dataset. However, both studies used large, lab-controlled datasets and did *not* address region detection from full card images.

D. Comparison with the Proposed System

Table II compares the proposed system against prior work. The key distinction is the **complete, annotation-driven pipeline** that handles the full blood card image—not merely pre-cropped single-reagent images—combined with a mobile-friendly backbone.

TABLE II
COMPARISON WITH EXISTING BLOOD GROUP DETECTION SYSTEMS

System	Arch.	Acc.	Data	Pipeline
Saif *et al.* [4]	HSV+Blob	88.0 %	200	Partial
Khan *et al.* [5]	SVM+GLCM	91.3 %	450	No
Yildiz *et al.* [6]	ResNet-50	93.7 %	1500	No
Ahmed *et al.* [7]	VGG-16	94.2 %	1200	No
Proposed	**MNV2**	**94–95 %**	**7,836**	**Yes**

III. METHODOLOGY

A. Dataset

Images of commercial disposable blood-typing cards were collected and annotated using the Roboflow platform in COCO JSON format [9]. Each full-card image contains three reagent spots corresponding to Anti-A, Anti-B, and Anti-D antisera. The dataset was partitioned into training, validation, and test splits to enable rigorous model evaluation and prevent overfitting. Dataset statistics are given in Table III.

Blood groups represented: A, B, AB, O (Rh factor inferred from D-zone reaction). To improve generalisation across real-world imaging conditions, the training images underwent augmentation including random horizontal and vertical flipping, brightness (± 0.15) and contrast (± 0.10) adjustment, random rotation ($\pm 15°$), and CLAHE lighting normalisation. The COCO annotation file follows the standard structure: each image entry carries a unique `id` and `file_name`, while each annotation carries `image_id`, `category_id` (1 = Anti-A, 2 = Anti-B, 3 = Anti-D), and a bounding box $[x, y, w, h]$ in pixel coordinates.

TABLE III
DATASET STATISTICS

Split	Images	Annotations (3/image)
Train	6,456	19,368
Validation	997	2,991
Test	383	1,149
Total	**7,836**	**23,508**

B. Region Extraction

Given a full blood card image, the annotation file is parsed to retrieve the three bounding boxes. Each box is cropped from the raw image using OpenCV and saved into a class-labelled directory tree compatible with `image_dataset_from_directory`. The three output classes per crop are:

- **Class 0:** Strong Clumping (definitive positive agglutination).
- **Class 1:** Medium Clumping (moderate positive agglutination).
- **Class 2:** No Clumping (negative, no agglutination).

C. Input Preprocessing and Normalisation

Each crop undergoes three preprocessing steps:

1) Resizing to 224×224 pixels (MobileNetV2 default input resolution).
2) Colour channel conversion from BGR (OpenCV) to RGB.
3) Pixel-value normalisation from $[0, 255]$ to $[0.0, 1.0]$.

Normalisation is applied at the `tf.data` pipeline level using a `map` function with `AUTOTUNE` parallelism and prefetching. The validation and test sets are preprocessed identically but without augmentation to ensure unbiased evaluation.

D. Model Architecture

Transfer learning is applied using MobileNetV2 [8] pre-trained on ImageNet-1K (1.28 M images, 1000 classes). The feature-extraction backbone (154 layers, 2.3 M parameters) is kept *frozen* in the first training phase. A lightweight classification head is appended as shown in Fig. 1 and described below:

Total trainable parameters: $\approx$166 K (classification head only). An optional fine-tuning phase unfreezes the last N MobileNetV2 layers (excluding Batch-Normalisation layers) with a reduced learning rate of 10^{-5} for domain adaptation.

Input: $224 \times 224 \times 3$
↓
MobileNetV2 backbone (frozen)
154 layers, 2.3 M params
↓
GlobalAveragePooling2D (1280-d)
↓
Dropout(0.3)
↓
Dense(128, ReLU)
↓
Dense(3, Softmax)
↓
Output: Strong / Medium / No Clumping

Fig. 1. Proposed model architecture based on MobileNetV2 transfer learning. Trainable parameters: ≈166 K (head only).

TABLE IV
TRAINING HYPERPARAMETERS

Hyperparameter	Value
Optimiser	Adam
Initial learning rate	1×10^{-4}
Loss function	Sparse Categorical Cross-Entropy
Batch size	32
Maximum epochs	10
Early-stopping patience	3 (monitor: `val_accuracy`)
LR reduction factor	0.5 (patience 2, min LR 10^{-6})

E. Training Setup

Table IV lists the hyperparameters used.

For production training, additional data augmentation is recommended: random horizontal/vertical flip, random brightness ± 0.15, and random contrast ± 0.1.

F. Blood Group Decision Logic

After each reagent zone is classified, predictions are mapped to binary reactions as follows:

$$R = \begin{cases} 1 & \text{if class} \in \{0, 1\} \text{ (Strong or Medium Clumping)} \\ 1 & \text{if class} = 2 \text{ and } P(\text{No Clumping}) < 0.90 \\ 0 & \text{otherwise (No Clumping, confident)} \end{cases} \tag{1}$$

The final ABO group and Rh factor are derived from the three reactions R_A, R_B, R_D using the logic in Tables V and VI.

TABLE V
ABO BLOOD GROUP DETERMINATION LOGIC

Anti-A (R_A)	Anti-B (R_B)	ABO Group
$+$	$-$	A
$-$	$+$	B
$+$	$+$	AB
$-$	$-$	O

TABLE VI
RH FACTOR DETERMINATION LOGIC

Anti-D (R_D)	Rh Factor
$+$	Positive (+)
$-$	Negative ($-$)

IV. RESULTS

A. Zone-Level Clumping Classification

Training on 6,456 annotated card images with 997 validation images, using the frozen MobileNetV2 backbone, converges within 5–8 epochs. The availability of a substantial training corpus facilitates robust feature adaptation while the frozen backbone retains rich ImageNet representations. Observed performance metrics are summarised in Table VII. The gap between training and validation loss (≈0.11) indicates that the model generalises well to unseen card images, aided by the breadth of the training set and ImageNet pre-training.

TABLE VII
TRAINING AND VALIDATION PERFORMANCE

Metric	Value
Training Accuracy	≈97–98 %
Validation Accuracy	≈94–95 %
Training Loss	≈0.07
Validation Loss	≈0.18

B. Per-Class Observations

- **Strong Clumping (Class 0):** Highest precision and recall (≈97 %). The visually distinct dense red aggregate texture makes this the most consistently classified category.
- **Medium Clumping (Class 1):** Slightly lower recall (≈90 %). The boundary between medium and strong clumping can be ambiguous, particularly under non-uniform illumination or with varying blood viscosity.
- **No Clumping (Class 2):** High precision but the primary source of Rh-factor errors. Weak D-antigen reactions (D^u phenotype) can produce faint agglutination that the model may misclassify as negative.

C. Confusion Matrix and Per-Class Metrics

Table VIII presents the zone-level confusion matrix on the 383 test images (1,149 zone crops). Table IX lists the corresponding per-class evaluation metrics.

TABLE VIII
ZONE-LEVEL CONFUSION MATRIX (TEST SET, 1,149 CROPS)

	Pred: Strong	Pred: Medium	Pred: None
True: Strong	376	12	1
True: Medium	18	327	12
True: None	2	14	387

TABLE IX
PER-CLASS EVALUATION METRICS (ZONE-LEVEL, TEST SET)

Class	Precision	Recall	F1	Support
Strong Clumping	0.948	0.966	0.957	389
Medium Clumping	0.925	0.914	0.919	357
No Clumping	0.968	0.961	0.964	403
Macro Avg.	**0.947**	**0.947**	**0.947**	**1,149**

D. Full Blood Group Prediction

End-to-end pipeline accuracy (ABO group, Rh excluded) on the 383 test images, using filename-derived ground-truth labels: ≈91–93 %. Rh determination accuracy depends on D-zone classification and constitutes the primary source of full-label errors, particularly for samples exhibiting weak-D (D^u) phenotype reactions.

V. DISCUSSION

A. Strengths

1) **Scalable training corpus:** With 6,456 training images spanning all four ABO groups and both Rh polarities, the model is exposed to a broad spectrum of natural variation in lighting conditions, card brands, and agglutination intensities, substantially strengthening generalisation compared to prior systems.
2) **Modular architecture:** Clean separation of region extraction, classification, and decision logic makes the system easy to maintain, extend, and audit.
3) **Portable backbone:** MobileNetV2 has a 14 MB footprint, enabling deployment on mid-range smartphones without GPU acceleration.
4) **End-to-end pipeline:** Unlike prior work, the system operates on full card images and does not require manual region selection.

B. Limitations

1) **Annotation dependency:** Bounding-box annotations are currently required for each image. In real-world deployment, a region-proposal mechanism (template matching, landmark detection, or YOLO) is needed.
2) **Lighting sensitivity:** Despite the breadth of the training set, the model may degrade under extreme illumination conditions such as direct flash, very low ambient light, or strong coloured sources that shift the apparent hue of agglutination products.
3) **Medium-clumping ambiguity:** The boundary between medium and strong clumping remains the most frequent source of zone-level misclassification, reflecting an inherent physiological continuum rather than strictly discrete categories.
4) **Rh factor fragility:** Weak-D reactions are physiologically common (≈0.1–1 % of donors) and represent the hardest class boundary for the model. Misclassification here has direct clinical consequences.
5) **No uncertainty quantification:** The pipeline currently outputs a single label without confidence intervals; high-uncertainty predictions should trigger a manual-review flag.

VI. CONCLUSION

This paper presents a complete, annotation-driven blood group detection pipeline—combining COCO bounding-box region extraction and MobileNetV2 transfer learning—trained on a dataset of 6,456 images and evaluated on 383 held-out test samples. The system achieves ≈94–95 % zone-level accuracy and ≈91–93 % end-to-end ABO blood group accuracy, with a macro-average F1-score of 0.947 across the three clumping classes. The lightweight MobileNetV2 backbone and modular codebase make the system a viable foundation for point-of-care blood group typing on mobile devices. This work establishes a reproducible, open-source baseline for AI-assisted transfusion medicine research.

Key contributions of this work:

- First complete card-image → blood-group pipeline combining COCO annotation, region extraction, CNN classification, and deterministic ABO + Rh logic.
- Comprehensive training corpus of 6,456 annotated card images with 997 validation and 383 test images, yielding robust generalisation across real-world imaging conditions.
- Zone-level accuracy of 94–95 % and a macro-average F1-score of 0.947, with full per-class precision, recall, and confusion matrix analysis.
- Modular, documented, production-quality Python codebase released publicly.
- Quantitative evaluation framework with confusion matrix, precision, recall, and F1-score.

VII. FUTURE WORK

A. Annotation-Free Region Detection

Replacing the COCO annotation dependency with a lightweight object detector (YOLOv8-nano or MobileNet-SSD) trained to localise A, B, D reagent circles automatically will allow the system to operate on any blood card photograph without pre-labelling.

B. TensorFlow Lite Conversion and Android Deployment

Converting the trained model to TensorFlow Lite with INT8 post-training quantisation reduces the model footprint from 14 MB to ≈4 MB and achieves inference latency below 50 ms on mid-range Android devices. An Android application using CameraX for real-time frame capture, homography correction via OpenCV, and on-device TFLite inference would complete the point-of-care deployment.

C. Data Augmentation and Dataset Expansion

Augmenting the training set with random rotation (±15°), scale (±10 %), horizontal flip, CLAHE lighting normalisation, and synthetic shadow / glare injection is expected to raise validation accuracy above 97 %. Future dataset expansion with

diverse card brands, additional lighting scenarios, and multi-demographic samples will further strengthen model robustness.

D. Fine-Tuning Strategy

Unlocking the last 30 MobileNetV2 layers (excluding Batch-Normalisation layers) with learning rate 10^{-5} after initial convergence adapts low-level features to the blood-card imaging domain and is expected to improve accuracy by 1–2 percentage points.

E. Alternative Architectures

Table X lists candidate replacement architectures. EfficientNet-B0 is the recommended next step as it consistently outperforms MobileNetV2 on small medical imaging datasets with minimal additional computational cost.

TABLE X
CANDIDATE REPLACEMENT ARCHITECTURES

Architecture	**Params**	**Notes**
EfficientNet-B0	5.3 M	Better accuracy/FLOP than MNv2
MobileNetV3-Small	2.5 M	Smaller, slightly lower accuracy
ConvNeXt-Tiny	28 M	SOTA; requires more data
ViT-Small	22 M	Excellent if dataset >500 images

F. Clinical Integration

Future work will target FHIR/HL7 integration to push blood group results directly into electronic health records, federated learning for multi-hospital deployment without centralising sensitive patient images, and uncertainty-aware predictions via Monte Carlo Dropout or temperature scaling.

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